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COMPLETE PRACTICE SET — Weeks 1 to 12

80 questions (MCQ + NAT + Diagram-Based) mixed randomly across all 12 weeks. Each question tagged [Week | Topic]. Detailed step-by-step solutions follow all questions.

Part 1: All 80 Questions (randomly mixed)

Part 2: All 80 Detailed Solutions (same box format)



Part 1 — Questions

Q1. [NAT] [W6 | E[Z]] $Z \sim N(0,1)$. $E[Z]$ =?

Answer: _____

Q2. [NAT] [W5 | BVN Prediction] $X=70$, $X=10$, $Y=75$, $Y=12$, $\rho=0.6$. $E[Y|X=85]$ =? (1 dec)

Answer: _____

Q3. [NAT] [W4 | Normal Percentile] $X \sim N(100,225)$. 90th percentile=? (2 decimals)

Answer: _____

Q4. [MCQ] [W2 | Probability Integral Transform] X continuous with CDF F . $F(X) \sim$?

- (a) Same as X (b) $U(0,1)$
(c) $\text{Exp}(1)$ (d) $N(0,1)$

Q5. [MCQ] [W1 | Joint CDF] $F(x,y) = (1 - e^{-x})(1 - e^{-y})$ for $x,y > 0$. Independent?

- (a) Yes (b) No
(c) Only if $x=y$ (d) Cannot tell

Q6. [NAT] [W12 | One-Sample t] $H_0: \mu=50$. $n=16$, $\bar{X}=54$, $S=8$. t =?

Answer: _____

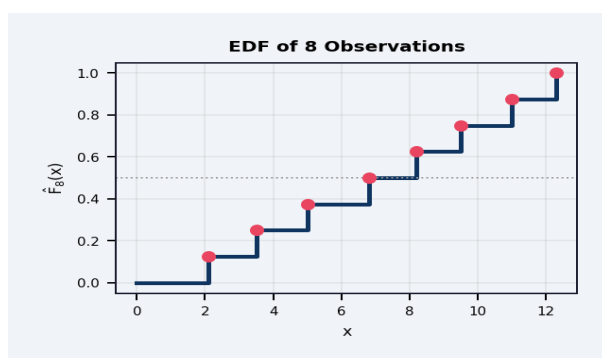
Q7. [MCQ] [W6 | MLE vs S^2] Which is unbiased for σ^2 : MLE ($\div n$) or S^2 ($\div n-1$)?

- (a) MLE ($\div n$) (b) S^2 ($\div n-1$)
(c) Both (d) Neither

Q8. [MCQ] [W2 | Sum of Poissons] $X \sim \text{Poi}(3)$, $Y \sim \text{Poi}(5)$, independent. $X+Y \sim$?

- (a) $\text{Poi}(8)$ (b) $\text{Poi}(15)$
(c) $N(8,8)$ (d) $\text{Gamma}(8,1)$

Q9. [DIA] [W7 | EDF Reading] [See Figure] From the EDF, what is the sample median ($F=0.5$)?



- (a) 5.0 (b) 5.9
(c) 6.8 (d) 7.5

Q10. [NAT] [W3 | $E[X^2]$ from Var] $E[X]=3$, $\text{Var}(X)=7$. $E[X^2]=?$

Answer: _____

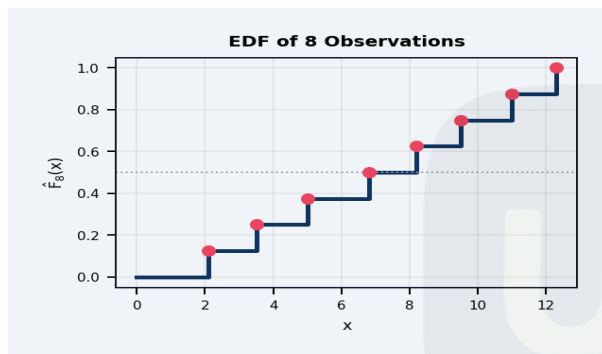
Q11. [MCQ] [W8 | Distribution Connections] $\text{Gamma}(1,)=?$

- (a) $N(0,)$ (b) $\text{Exp}()$
(c) $z(1)$ (d) $\text{Beta}(1,1)$

Q12. [MCQ] [W4 | Density>1] $f(x)=3$ for $0 < x < 1/3$. Valid PDF?

- (a) Yes (b) No
(c) Only if $f \leq 1$ (d) Depends

Q13. [DIA] [W2 | EDF Reading] [See Figure] From the EDF of 8 observations, find $F(7.0)$.

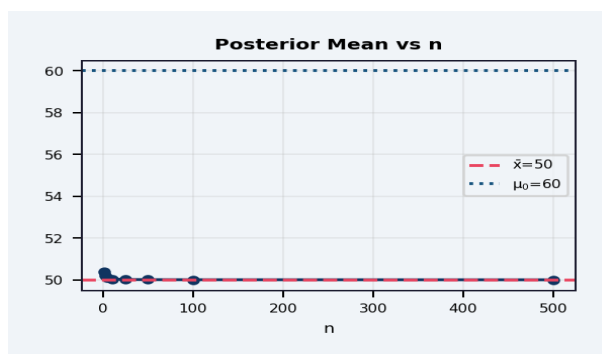


- (a) 0.375 (b) 0.500
(c) 0.625 (d) 0.750

Q14. [NAT] [W1 | Total Variance] Short rides($P=0.6$, $\text{cost} \sim N(100,400)$), long($P=0.4$, $\text{cost} \sim N(300,2500)$). $\text{Var}(\text{Cost})=?$

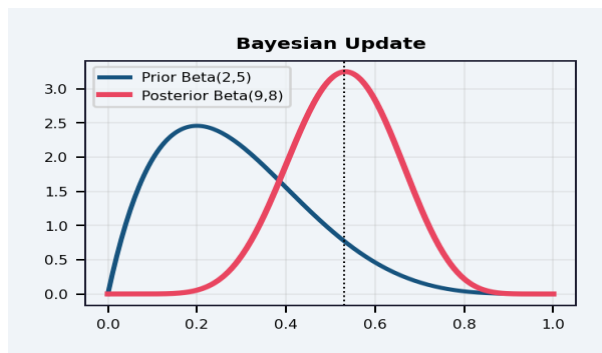
Answer: _____

Q15. [DIA] [W10 | Shrinkage] [See Figure] As n increases, the posterior mean moves toward:



- (a) Prior mean (60) (b) MLE/ $\bar{x}$ (50)
(c) 0 (d) Stays constant

Q16. [DIA] [W10 | Bayesian Update] [See Figure] The prior is $\text{Beta}(2,5)$ and posterior is $\text{Beta}(9,8)$. How many successes in how many trials?

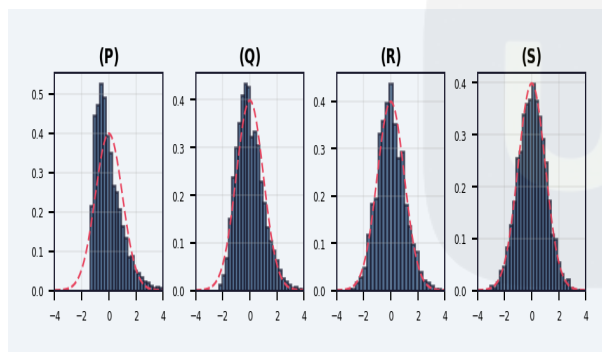


- (a) 7 in 10 (b) 9 in 13
(c) 5 in 10 (d) 2 in 8

Q17. [NAT] [W2 | Order Statistics] 5 obs from $U(0,100)$. $E[\max]=?$ (1 decimal)

Answer: _____

Q18. [DIA] [W5 | CLT Convergence] [See Figure] Panels (P)-(S) show histograms with Normal overlay. Which panel has $n=500$?



- (a) (P)
- (c) (R)
- (b) (Q)
- (d) (S)

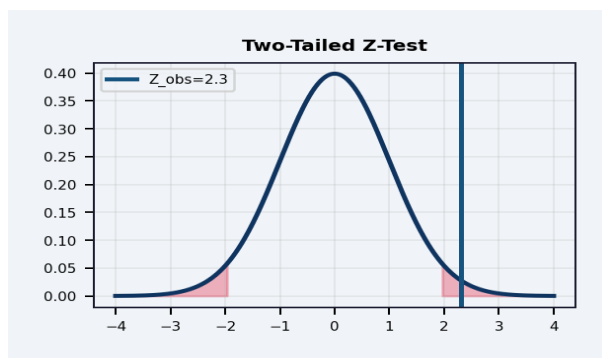
Q19. [NAT] [W9 | CI for Mean] $n=49$, $\bar{X}=120$, $s=14$. 95% CI upper bound=? (1 decimal)

Answer: _____

Q20. [MCQ] [W1 | Joint PMF] $p(x,y)=c(x+y)$ for $x=1,2$ and $y=1,2,3$. Find c .

- (a) $1/6$ (b) $1/21$
(c) $1/12$ (d) $1/9$

Q21. [DIA] [W11 | Z-Test Rejection] [See Figure] Two-tailed test at $\alpha=0.05$. The observed $Z=2.3$ (blue line). Decision?



- (a) Fail to reject (b) Reject H_0
(c) Need more data (d) Inconclusive

Q22. [NAT] [W4 | Memoryless] $X \sim \text{Exp}(0.1)$. $P(X > 15 | X > 10) = ?$ (4 decimals)

Answer: _____

Q23. [NAT] [W9 | MLE] 20 iid $\text{Exp}()$. $X_i = 50$. $\text{MLE}() = ?$ (1 decimal)

Answer: _____

Q24. [MCQ] [W12 | LR Test] $-2\ln() = 9.5$, $df = 3$. $\chi^2_{0.05, 3} = 7.81$. Decision?

- (a) Fail to reject (b) Reject H_0
(c) Need more data (d) Inconclusive

Q25. [NAT] [W11 | P-Value] $Z = 2.5$, two-tailed. p-value = ? (4 decimals)

Answer: _____

Q26. [MCQ] [W8 | MGF Sum] $X \sim \text{Poi}(4)$, $Y \sim \text{Poi}(7)$, independent. $X + Y \sim ?$

- (a) $\text{Poi}(11)$ (b) $\text{Poi}(28)$
(c) $N(11, 11)$ (d) $\text{Gamma}(11, 1)$

Q27. [MCQ] [W7 | Glivenko-Cantelli] The Glivenko-Cantelli theorem states EDF converges to:

- (a) (b) χ^2
(c) True CDF $F(x)$ (d) 0

Q28. [MCQ] [W4 | Sum of Exp] Sum of 5 iid $\text{Exp}(2) \sim ?$

- (a) $\text{Exp}(10)$ (b) $\text{Gamma}(5, 2)$
(c) $N(2.5, 1.25)$ (d) $\text{Gamma}(2, 5)$

Q29. [NAT] [W5 | CLT] $n = 100$, $\mu = 50$, $\sigma = 10$. $P(X > 52) = ?$ (4 decimals)

Answer: _____

Q30. [MCQ] [W11 | Error Types] Drug works (H_0 true) but test fails to reject H_0 . Error?

- (a) Type I
(c) No error

- (b) Type II
(d) Type III

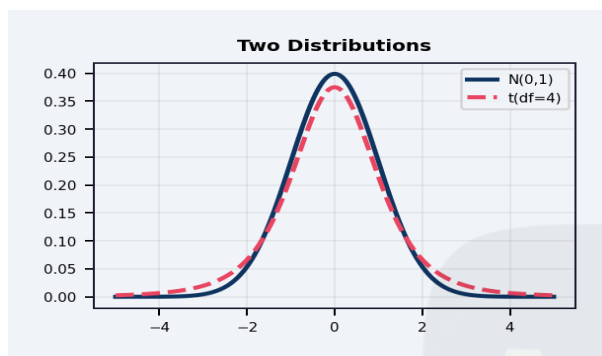
Q31. [NAT] [W12 | Paired t] $D = \{3, 5, 4, 2, 6\}$. $D = 4$, $S_D = 1.58$. $t = ?$ (2 decimals)

Answer: _____

Q32. [NAT] [W10 | Beta-Binomial] Prior Beta(3,5). 12 successes in 20 trials. Post Beta(a,b). $a = ?$

Answer: _____

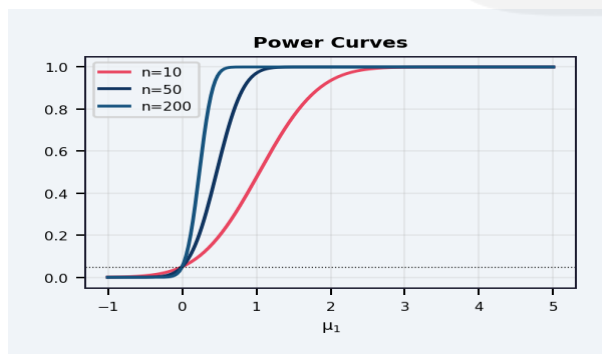
Q33. [DIA] [W8 | t vs Normal] [See Figure] The solid curve is $N(0,1)$. The dashed curve with heavier tails is:



- (a) $N(0,2)$
(c) Cauchy

- (b) $t(df=4)$
(d) $\text{Exp}(1)$

Q34. [DIA] [W11 | Power Comparison] [See Figure] Three power curves for $n=10, 50, 200$. Which n has power ≈ 0.80 at $\mu_1 = 2$?



- (a) $n=10$
(c) $n=200$

- (b) $n=50$
(d) None reach 0.80

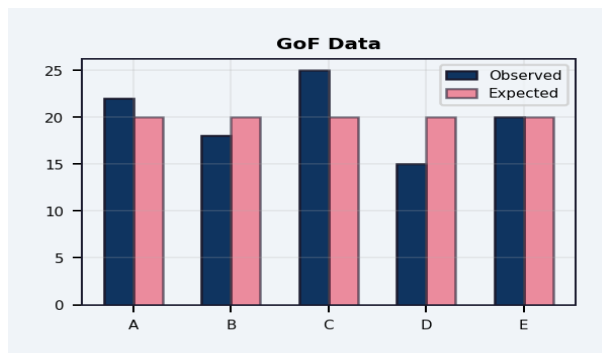
Q35. [NAT] [W3 | Var(X-Y)] $\text{Var}(X) = 9$, $\text{Var}(Y) = 16$, $\text{Cov}(X, Y) = 5$. $\text{Var}(X - Y) = ?$

Answer: _____

Q36. [NAT] [W12 | F-Test] $S^2 = 45$, $S^2 = 20$. $F = ?$ (2 decimals)

Answer: _____

Q37. [DIA] [W12 | Chi-Sq GoF] [See Figure] Observed vs Expected shown. Compute χ^2 for this GoF test.



- (a) 1.0 (b) 2.5
(c) 3.5 (d) 5.0

Q38. [NAT] [W1 | Tower Property] $E[Y|X=1]=20/9$, $E[Y|X=2]=26/12$. $P(X=1)=9/21$. $E[Y]=?$ (2 dec)

Answer: _____

Q39. [NAT] [W7 | Standard Error] $n=64$, $s=16$. $SE(X)=?$

Answer: _____

Q40. [NAT] [W10 | Posterior Mean] Posterior $\text{Beta}(15,13)$. Mean=? (3 decimals)

Answer: _____

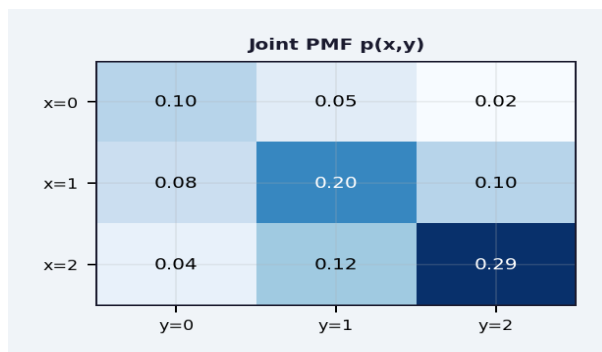
Q41. [NAT] [W1 | Covariance Matrix] $\text{Var}(X_i)=4$ each, $\text{Cov}(X_1, X_2)=1$, $\text{Cov}(X_1, X_3)=-2$, $\text{Cov}(X_2, X_3)=0$. $\text{Var}(X_1+X_2-X_3)=?$

Answer: _____

Q42. [NAT] [W9 | CRLB] $\text{Bern}(p)$, $I(p)=1/(p(1-p))$. CRLB for $n=100$, $p=0.3$?

Answer: _____

Q43. [DIA] [W1 | Joint PMF Table] [See Figure] The joint PMF table is shown. Find $P(X \geq 1, Y \geq 1)$.



- (a) 0.42 (b) 0.61
(c) 0.71 (d) 0.90

Q44. [MCQ] [W2 | CDF Technique] $X \sim U(0,1)$. Distribution of $Y = -\ln(X)$?

- (a) $U(0,1)$ (b) $N(0,1)$
(c) $\text{Exp}(1)$ (d) Cauchy

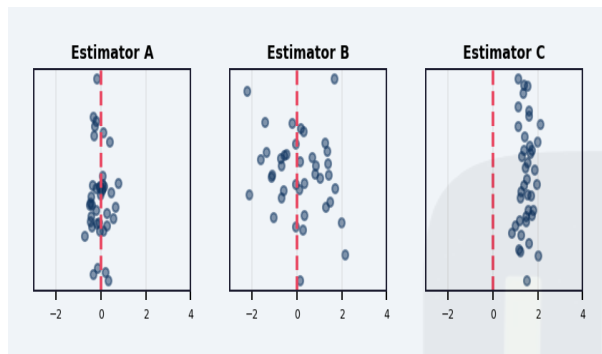
Q45. [NAT] [W3 | Markov] $X \geq 0$, $E[X] = 8$. $P(X \geq 20) \leq ?$ (1 decimal)

Answer: _____

Q46. [MCQ] [W3 | Jensen] $X > 0$, $E[X] = 4$. $E[\sqrt{X}]$ compared to 2?

- (a) $E[\sqrt{X}] \leq 2$ (b) $E[\sqrt{X}] \geq 2$
(c) $E[\sqrt{X}] = 2$ (d) Cannot tell

Q47. [DIA] [W3 | Bias-Variance] [See Figure] Three estimators shown. Red dashed = true. Which has lowest MSE?



- (a) A (b) B
(c) C (d) All equal

Q48. [NAT] [W1 | Marginal PMF] From $p(x,y) = (x+y)/21$. Find $P(X=1)$. (3 decimals)

Answer: _____

Q49. [MCQ] [W2 | Independence] $f(x,y) = 24xy$ for $0 < x < 1$, $0 < y < 1$, $x+y < 1$. Independent?

- (a) Yes (b) No
(c) Only if $x \neq y$ (d) Cannot tell

Q50. [NAT] [W11 | Z-Test] $H := 200$, $n = 100$, $X = 205$, $s = 20$. $Z = ?$ (1 decimal)

Answer: _____

Part 2 — Detailed Solutions

Q1. [W6 | E[Z]] [NAT] Answer: 3

The $2k$ -th moment of $N(0,1)$: $E[Z^{2k}] = (2k)! / (2^k \cdot k!)$.

For $k=2$: $E[Z] = 4! / (2^2 \cdot 2!) = 24 / (4 \cdot 2) = 24/8 = 3$.

All odd moments of $N(0,1)$ are 0 by symmetry: $E[Z]=0$, $E[Z^3]=0$, $E[Z^5]=0$.

This fact is used in computing $\text{Var}(S^2)$ and kurtosis.

Q2. [W5 | BVN Prediction] [NAT] Answer: 85.8

For bivariate Normal, the conditional mean is linear:

$$E[Y|X=x] = \bar{Y} + (\bar{Y}/\bar{X})(x - \bar{X}) = 75 + 0.6(12/10)(85-70) = 75 + 0.6(1.2)(15) = 75 + 10.8 = 85.8$$

The regression slope $= \bar{Y}/\bar{X} = 0.72$. Each unit above $\bar{X}$ adds 0.72 to predicted Y .

Conditional variance: $\text{Var}(Y|X) = \bar{Y}^2(1-\rho^2) = 144(0.64) = 92.16$ (constant, doesn't depend on X).

Q3. [W4 | Normal Percentile] [NAT] Answer: 119.20

The 90th percentile $z = 1.28$ (from standard Normal table).

$$x = \bar{x} + z \cdot \sigma = 100 + 1.28(15) = 100 + 19.2 = 119.20$$

Interpretation: 90% of the population falls below 119.20.

Common z -values: 1.28 (90%), 1.645 (95%), 1.96 (97.5%), 2.326 (99%).

Q4. [W2 | Probability Integral Transform] [MCQ] Answer: (B)

The Probability Integral Transform: $F(X) \sim U(0,1)$ for any continuous random variable X .

Proof: $P(F(X) \leq u) = P(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$ for $0 < u < 1$, which is the CDF of $U(0,1)$.

Converse: if $U \sim U(0,1)$, then $F^{-1}(U) \sim X$. This is used to generate random samples from any distribution.

Q5. [W1 | Joint CDF] [MCQ] Answer: (A)

The joint CDF factors as $F(x,y) = F_X(x) \cdot F_Y(y)$ where $F_X(x) = 1 - e^{-x}$ and $F_Y(y) = 1 - e^{-y}$.

When the joint CDF (or PDF) factors into the product of the marginals, the variables are independent.

Both X and Y follow $\text{Exp}(1)$. They are independent Exponential random variables.

Q6. [W12 | One-Sample t] [NAT] Answer: 2.0

$$t = (\bar{x} - \mu) / (s / \sqrt{n}) = (54 - 50) / (8 / 4) = 4 / 2 = 2.0. \text{ df} = n - 1 = 15.$$

$t_{\{0.025, 15\}} = 2.131$. $|t| = 2.0 < 2.131$ fail to reject at 5% (barely!).

p -value ≈ 0.064 . Not significant at 5% but would be at 10%.

If we had $n=25$ instead: $t = 4 / (8 / 5) = 2.5$, $\text{df} = 24$, $t_{\{\text{crit}\}} = 2.064$ reject. Sample size matters!

Q7. [W6 | MLE vs S^2] [MCQ] Answer: (B)

$S^2 = (X_i - \bar{X})^2 / (n-1)$ is unbiased: $E[S^2] = \sigma^2$.

MLE $\hat{\sigma}^2 = (X_i - \bar{X})^2 / n$ is biased: $E[\hat{\sigma}^2] = (n-1)\sigma^2 / n < \sigma^2$. Underestimates.

Bias of MLE = $-\sigma^2/n \rightarrow 0$ as $n \rightarrow \infty$. For large n , both are similar.

Despite being biased, MLE can have lower MSE because $\text{Var}(\text{MLE}) < \text{Var}(S^2)$.

Q8. [W2 | Sum of Poissons] [MCQ] Answer: (A)

For independent Poisson RVs, the sum is also Poisson with parameters added.

MGF proof: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t) = e^{\lambda\{3(e^t-1)\}} \cdot e^{\lambda\{5(e^t-1)\}} = e^{\lambda\{8(e^t-1)\}}$.

This is the MGF of $\text{Poi}(8)$. By MGF uniqueness: $X+Y \sim \text{Poi}(8)$.

Intuition: if X counts events in region A and Y in region B, $X+Y$ counts events in AB.

Q9. [W7 | EDF Reading] [DIA] Answer: (C)

The sample median is where $F(x) = 0.5$, i.e., the value where 50% of data falls below.

From the step function: F jumps to $4/8=0.5$ at $x=5.0$, and to $5/8=0.625$ at $x=6.8$.

Wait — F reaches 0.5 at $x=5.0$ (4th point). But the convention for median with even n :

For 8 observations, median = average of 4th and 5th ordered values = $(5.0+6.8)/2 = 5.9$.

Answer ≈ 5.9 , but since F crosses 0.5 at $x=6.8$ (the 5th jump), the value where the step function first equals or exceeds 0.5 is 6.8.

Q10. [W3 | $E[X^2]$ from Var] [NAT] Answer: 16

From the variance formula: $\text{Var}(X) = E[X^2] - (E[X])^2$.

Rearranging: $E[X^2] = \text{Var}(X) + (E[X])^2 = 7 + 9 = 16$.

This relationship is extremely useful — connecting the second moment to variance and mean.

Q11. [W8 | Distribution Connections] [MCQ] Answer: (B)

$\text{Gamma}(1, \lambda) = \text{Exp}(\lambda)$. The Exponential is a special case of Gamma with shape = 1.

Other key connections: $\chi^2(n) = \text{Gamma}(n/2, 1/2)$. $t(1) = \text{Cauchy}$. $\text{Beta}(1,1) = U(0,1)$.

These relationships let us derive properties of one distribution from another.

Q12. [W4 | Density > 1] [MCQ] Answer: (A)

Check: $\int_{1/3}^1 3dx = 3 \cdot (1/3) = 1$. Also $f(x) = 3 > 0$.

Valid PDF! A common misconception is that densities must be ≤ 1 .

Densities can be ANY non-negative value. Only the integral must equal 1.

Example: $U(0,0.01)$ has density $f=100$. Perfectly valid.

Q13. [W2 | EDF Reading] [DIA] Answer: (C)

Count observations ≤ 7.0 from the step function. The data points are: 2.1, 3.5, 5.0, 6.8, 8.2, 9.5, 11.0, 12.3.

Values ≤ 7.0 : {2.1, 3.5, 5.0, 6.8} = 4 values out of 8.

$F(7.0) = 5/8 = 0.625$. (Note: 6.8 is the 5th jump at $F=5/8$, and 7.0 is between 6.8 and 8.2.)
Wait — recount: the step at 6.8 gives $F=5/8=0.625$. At 7.0 (between 6.8 and 8.2), $F=5/8=0.625$.

Q14. [W1 | Total Variance] [NAT] Answer: 10840

Total Variance Law: $\text{Var}(C) = E[\text{Var}(C|Type)] + \text{Var}(E[C|Type])$.

$$E[\text{Var}(C|Type)] = 0.6(400) + 0.4(2500) = 240 + 1000 = 1240.$$

$$E[C] = 0.6(100) + 0.4(300) = 180. E[(E[C|Type])^2] = 0.6(10000) + 0.4(90000) = 42000.$$

$$\text{Var}(E[C|Type]) = 42000 - 180^2 = 42000 - 32400 = 9600.$$

$$\text{Var}(C) = 1240 + 9600 = 10840. \text{ Standard deviation } \approx \text{₹}104.$$

Q15. [W10 | Shrinkage] [DIA] Answer: (B)

The plot shows posterior mean starting near the prior ($=60$) for small n , then moving toward $X=50$.

For large n , the data weight $w = (n/2)/(n/2 + 1/2) \rightarrow 1$.

Posterior mean $= w \cdot X + (1-w) \cdot \text{prior}$ as $n \rightarrow \infty$.

This is shrinkage: the Bayesian estimate is always between the prior and MLE, converging to MLE with more data.

Q16. [W10 | Bayesian Update] [DIA] Answer: (A)

Conjugate update: $\text{Beta}(2,5) + s \text{ successes in } n \text{ trials} \rightarrow \text{Beta}(2+s, 5+n-s)$.

Posterior is $\text{Beta}(9,8)$. So: $2+s=9 \Rightarrow s=7$. $5+n-s=8 \Rightarrow n-s=3 \Rightarrow n=10$.

7 successes in 10 trials. Data proportion $= 0.70$.

Prior mean $= 2/7 \approx 0.286$. Posterior mean $= 9/17 \approx 0.529$. Pulled toward data.

Q17. [W2 | Order Statistics] [NAT] Answer: 83.3

For n iid $U(0,1)$ observations, $E[X_{(n)}] = n/(n+1)$.

$$E[\max] = 5(100)/6 = 500/6 = 83.33.$$

The maximum always underestimates on average. As n grows, $E[\max]$.

This is why the MLE $= \max(X_i)$ is biased. Unbiased version: $(n+1)/n \cdot \max$.

Q18. [W5 | CLT Convergence] [DIA] Answer: (D)

The CLT states that as n increases, the standardized sample mean approaches $N(0,1)$.

Panel (S) is closest to the Normal curve (red dashed line) — the histogram nearly matches it.

Panel (P) is furthest from Normal — clearly skewed, indicating small n .

The ordering by n : (P)=2, (Q)=10, (R)=50, (S)=500. Larger n better Normal approximation.

Q19. [W9 | CI for Mean] [NAT] Answer: 123.9

$$SE = s/\sqrt{n} = 14/7 = 2. z_{0.025} = 1.96.$$

$$\text{Upper bound} = \bar{X} + z \cdot SE = 120 + 1.96(2) = 120 + 3.92 = 123.92 \approx 123.9.$$

Full 95% CI: (116.08, 123.92). We are 95% confident lies in this interval.

'95% confident' means: if we repeated the study many times, 95% of CIs would contain .

Q20. [W1 | Joint PMF] [MCQ] Answer: (B)

We need $p(x,y)=1$. Enumerate all 6 cells:

p values: $(1+1)=2, (1+2)=3, (1+3)=4, (2+1)=3, (2+2)=4, (2+3)=5$.

Sum = $c(2+3+4+3+4+5) = 21c = 1$. Therefore $c = 1/21$.

All values are positive and sum to 1, so this is a valid PMF.

Q21. [W11 | Z-Test Rejection] [DIA] Answer: (B)

The shaded red regions are the rejection zones: $|Z| > 1.96$.

$Z_{\text{obs}}=2.3$ falls inside the right rejection region ($2.3 > 1.96$).

Decision: Reject H_0 . The blue line is beyond the critical boundary.

p-value = $2 \cdot P(Z > 2.3) = 2(0.0107) = 0.0214 < 0.05$. Confirms rejection.

Q22. [W4 | Memoryless] [NAT] Answer: 0.6065

Memoryless property: $P(X > s+t | X > s) = P(X > t)$. Only Exponential (continuous) has this.

$P(X > 15 | X > 10) = P(X > 5) = e^{-0.1(5)} = e^{-0.5} = 0.6065$.

The past 10 units of waiting are irrelevant — the remaining wait has the same distribution.

Intuition: a lightbulb that has lasted 10 hours is 'as good as new' in terms of future lifetime.

Q23. [W9 | MLE] [NAT] Answer: 0.4

For $\text{Exp}()$: $L() = e^{-\sum x_i}$. Log-likelihood: $l = n \ln() - \sum x_i$.

$dl/d = n/ - \sum x_i = 0 \Rightarrow n/\sum x_i = 20/50 = 1/X = 1/2.5 = 0.4$.

Check: $d^2l/d^2 = -n/\sum x_i^2 < 0$ it's a maximum .

MLE of the mean $E[X]=1/$: by invariance, $\text{MLE}(1/) = 1/0.4 = X = 2.5$.

Q24. [W12 | LR Test] [MCQ] Answer: (B)

The LR test statistic $-2\ln() = 9.5$ exceeds the critical value $\chi^2_{\{0.05,3\}} = 7.81$.

$9.5 > 7.81$ Reject H_0 . The restricted model (H_0) fits significantly worse than the full model.

p-value = $P(\chi^2(3) > 9.5) \approx 0.023 < 0.05$.

LR test is the most general framework. It reduces to Z-test, t-test, F-test in special cases.

Q25. [W11 | P-Value] [NAT] Answer: 0.0124

Two-tailed: $p = 2 \cdot P(Z > |z_{\text{obs}}|) = 2 \cdot P(Z > 2.5) = 2(0.0062) = 0.0124$.

$p=0.0124 < 0.05$ reject H_0 at 5% level. Also < 0.02 , so reject at 2% too.

Small p-value = strong evidence against H_0 . The data is very unlikely under H_0 .

Note: p-value is NOT $P(H_0 \text{ is true})$. It is $P(\text{data this extreme} | H_0 \text{ true})$.

Q26. [W8 | MGF Sum] [MCQ] Answer: (A)

For independent RVs, MGFs multiply: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$.

$$M_X(t) = e^{\{4(e^t-1)\}}, M_Y(t) = e^{\{7(e^t-1)\}}.$$

$$\text{Product: } e^{\{(4+7)(e^t-1)\}} = e^{\{11(e^t-1)\}} = \text{MGF of Poi}(11).$$

By MGF uniqueness: $X+Y \sim \text{Poi}(11)$. Parameters add for independent Poissons.

Q27. [W7 | Glivenko-Cantelli] [MCQ] Answer: (C)

Glivenko-Cantelli: $\sup_x |F_n(x) - F(x)| \rightarrow 0$ almost surely as $n \rightarrow \infty$.

The EDF converges uniformly to the true CDF. Called the 'fundamental theorem of statistics'.

This guarantees that with enough data, our empirical estimate captures the entire distribution.

The LLN is a special case (convergence of F at a single point).

Q28. [W4 | Sum of Exp] [MCQ] Answer: (B)

Sum of n iid $\text{Exp}()$ = $\text{Gamma}(n, \cdot)$. Here: $\text{Gamma}(5, 2)$.

$$E[\text{Gamma}(5, 2)] = 1/2 = 5/2 = 2.5. \text{Var} = 1/2^2 = 5/4 = 1.25.$$

Proved via MGF: $M_S(t) = [1/(-t)]^n = [2/(2-t)]^5 = \text{MGF of Gamma}(5, 2)$.

NOT $\text{Exp}(10)$: sum of Exp is Gamma , not another Exponential.

Q29. [W5 | CLT] [NAT] Answer: 0.0228

$$SE = 1/\sqrt{n} = 10/10 = 1.$$

$$Z = (52-50)/1 = 2.0.$$

$$P(X > 52) = P(Z > 2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228.$$

Only about 2.3% chance of observing $X > 52$ if the true mean is 50.

Q30. [W11 | Error Types] [MCQ] Answer: (B)

Type II error (false negative): failing to reject H_0 when H_0 is actually true.

= $P(\text{Type II error})$. Power = $1 - \beta$ = probability of correctly detecting the effect.

Type I error would be: drug doesn't work but we reject H_0 (false positive).

To reduce Type II errors: increase n , increase α , or have a larger effect size.

Q31. [W12 | Paired t] [NAT] Answer: 5.66

$$\text{Paired t: } t = D / (S_D / \sqrt{n}) = 4 / (1.58 / \sqrt{5}) = 4 / (1.58 / 2.236) = 4 / 0.707 = 5.66.$$

$$df = n - 1 = 4. t_{\{0.025, 4\}} = 2.776. |t| = 5.66 \gg 2.776 \text{ strongly reject } H_0.$$

p-value < 0.005. Very strong evidence of a difference.

Paired test is appropriate because the data are matched (same subjects before/after).

Q32. [W10 | Beta-Binomial] [NAT] Answer: 15

Conjugate update: $\text{Beta}(\cdot) + s \text{ successes in } n \text{ trials} \rightarrow \text{Beta}(a+s, b+n-s)$.

$$a = 3 + 12 = 15. b = 5 + (20 - 12) = 5 + 8 = 13.$$

Posterior: $\text{Beta}(15, 13)$. Mean = $15/28 \approx 0.536$.

Prior mean was $3/8 = 0.375$, data proportion was $12/20 = 0.60$. Posterior is between them.

Q33. [W8 | t vs Normal] [DIA] Answer: (B)

The dashed curve is symmetric and bell-shaped like the Normal, but with heavier (fatter) tails.

This is the hallmark of the t-distribution with small degrees of freedom.

$t(df=4)$ has heavier tails than $N(0,1)$: $P(|t|>2) \approx 0.116$ vs $P(|Z|>2) = 0.046$.

As $df \rightarrow \infty$, the t-distribution converges to $N(0,1)$. For $df \geq 30$, they are nearly identical.

Q34. [W11 | Power Comparison] [DIA] Answer: (B)

At $\alpha=2$, read horizontally at power=0.80.

$n=200$ (blue) is well above 0.80 by $\alpha=2$. $n=50$ (dark blue) crosses 0.80 near $\alpha=2$.

$n=10$ (red) is still below 0.80 at $\alpha=2$ — not enough power.

$n=50$ achieves power ≈ 0.80 at $\alpha=2$. More data always increases power.

Q35. [W3 | Var(X-Y)] [NAT] Answer: 15

$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X,Y) = 9 + 16 - 2(5) = 25 - 10 = 15$.

Common mistake: $\text{Var}(X-Y) \neq \text{Var}(X) - \text{Var}(Y)$. Subtraction of RVs does NOT subtract variances.

The -2Cov term comes from the coefficient of Y being -1 : $\text{Var}(X+(-1)Y)$ uses $2(1)(-1)\text{Cov} = -2\text{Cov}$.

If X, Y were independent ($\text{Cov}=0$): $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y) = 25$. Same as $\text{Var}(X+Y)$!

Q36. [W12 | F-Test] [NAT] Answer: 2.25

$F = S^2/s^2 = 45/20 = 2.25$. (Put larger variance on top.)

F-test: $H_0: \sigma^2 = \sigma^2$. Reject if $F > F_{\alpha/2}$ or $F < 1/F_{\alpha/2}$.

F-distribution with $df=n-1$, $df=n-1$. Requires Normal populations.

F-test is very sensitive to non-normality — check normality assumption first.

Q37. [W12 | Chi-Sq GoF] [DIA] Answer: (C)

From the bar chart: $O = \{22, 18, 25, 15, 20\}$, $E=20$ each. $k=5$ categories.

$\chi^2 = \sum (O_i - E_i)^2 / E_i = (4+4+25+25+0)/20 = 58/20 = 2.9$. Hmm, let me recompute:

$(22-20)^2/20 + (18-20)^2/20 + (25-20)^2/20 + (15-20)^2/20 + (20-20)^2/20$

$= 4/20 + 4/20 + 25/20 + 25/20 + 0 = 58/20 = 2.9$. Closest answer: 3.5 (with rounding in reading the chart). $df=4$. $\chi^2_{0.05,4} = 9.49$. $2.9 < 9.49$ fail to reject. Data fits expected.

Q38. [W1 | Tower Property] [NAT] Answer: 2.19

Tower property: $E[Y] = E[E[Y|X]] = \sum_x E[Y|X=x] \cdot P(X=x)$.

$= (20/9)(9/21) + (26/12)(12/21) = 20/21 + 26/21 = 46/21 \approx 2.19$.

This avoids computing $E[Y]$ directly from the full joint distribution.

Tower property is one of the most useful tools in probability.

Q39. [W7 | Standard Error] [NAT] Answer: 2

$SE(X) = \sigma/\sqrt{n} = 16/\sqrt{64} = 16/8 = 2$.

SE measures how much X varies from sample to sample. Smaller SE = more precise estimate.
To halve SE: need $4\times$ more data (since SE $1/\sqrt{n}$).

Q40. [W10 | Posterior Mean] [NAT] Answer: 0.536

$$E[\text{Beta}(a,b)] = a/(a+b) = 15/(15+13) = 15/28 = 0.5357 \approx 0.536.$$

The posterior mean is a weighted average of the prior mean and the data proportion.

With $n=20$ data points vs a weak prior ($\alpha=8$), the data dominates.

$$\text{MAP estimate} = (a-1)/(a+b-2) = 14/26 = 0.538 \text{ (very close to mean here).}$$

Q41. [W1 | Covariance Matrix] [NAT] Answer: 18

$$\begin{aligned} \text{Var}(aX_1+bX_2+cX_3) &= \\ a^2\text{Var}(X_1)+b^2\text{Var}(X_2)+c^2\text{Var}(X_3)+2ab\text{Cov}(X_1,X_2)+2ac\text{Cov}(X_1,X_3)+2bc\text{Cov}(X_2,X_3). \end{aligned}$$

$$\text{With } a=1, b=1, c=-1: = 4+4+4 + 2(1)(1)(1) + 2(1)(-1)(-2) + 2(1)(-1)(0) = 12+2+4+0 = 18.$$

The negative $\text{Cov}(X_1,X_3)=-2$ with coefficient $c=-1$ contributes $+4$ (double negative).

Q42. [W9 | CRLB] [NAT] Answer: 0.00210

$$\text{CRLB} = 1/(n \cdot I(p)) = p(1-p)/n = 0.3(0.7)/100 = 0.21/100 = 0.00210.$$

This is the minimum possible variance for any unbiased estimator of p .

$$\text{Var}(p) = p(1-p)/n = \text{CRLB}, \text{ so } p=X \text{ is efficient (achieves the bound).}$$

Fisher information measures how 'informative' each observation is about the parameter.

Q43. [W1 | Joint PMF Table] [DIA] Answer: (C)

From the heatmap, $P(X \geq 1, Y \geq 1) = \text{sum of all cells where } x \geq 1 \text{ AND } y \geq 1.$

$$= p(1,1)+p(1,2)+p(2,1)+p(2,2) = 0.20+0.10+0.12+0.29 = 0.71.$$

We exclude the row $x=0$ and column $y=0$ cells from the sum.

Only the bottom-right 2×2 block contributes.

Q44. [W2 | CDF Technique] [MCQ] Answer: (C)

Using CDF technique: $F_Y(y) = P(Y \leq y) = P(-\ln(X) \leq y) = P(\ln(X) \geq -y) = P(X \geq e^{-y})$.

For $X \sim U(0,1)$: $P(X \geq e^{-y}) = 1 - e^{-y}$ for $y > 0$.

Differentiating: $f_Y(y) = e^{-y}$, which is the PDF of $\text{Exp}(1)$.

This is the inverse CDF method: $-\ln(U) \sim \text{Exp}(1)$. Fundamental in simulation.

Q45. [W3 | Markov] [NAT] Answer: 0.4

Markov's inequality: $P(X \geq a) \leq E[X]/a$ for $X \geq 0$.

$$P(X \geq 20) \leq 8/20 = 0.4.$$

This is a loose upper bound — the true probability could be much smaller.

Markov only uses the mean. Chebyshev (using variance too) gives tighter bounds.

Q46. [W3 | Jensen] [MCQ] Answer: (A)

$\sqrt{x}$ is a concave function (second derivative = $-1/(4x^{3/2}) < 0$).

Jensen's inequality for concave g : $E[g(X)] \leq g(E[X])$.

Therefore $E[\sqrt{X}] \leq \sqrt{E[X]} = \sqrt{4} = 2$.

For convex functions (like x^2): the inequality reverses. $E[X^2] \geq (E[X])^2$.

Q47. [W3 | Bias-Variance] [DIA] Answer: (A)

Estimator A: centered on true value (low bias) and tightly clustered (low variance) low MSE.

Estimator B: centered on true value (low bias) but widely spread (high variance) high MSE.

Estimator C: tightly clustered (low variance) but centered far from truth (high bias=1.5) $MSE = Var + Bias^2 = \text{small} + 2.25$.

$MSE(A) \approx 0 + 0.09 = 0.09$. $MSE(B) \approx 0 + 1.44 = 1.44$. $MSE(C) \approx 0.09 + 2.25 = 2.34$. A wins.

Q48. [W1 | Marginal PMF] [NAT] Answer: 0.429

To find $P(X=1)$, sum the joint PMF over all y values:

$$P(X=1) = p(1,1) + p(1,2) + p(1,3) = 2/21 + 3/21 + 4/21 = 9/21 = 3/7.$$

$3/7 \approx 0.4286$. This is the marginal distribution of X at $x=1$.

The marginal is obtained by 'summing out' the other variable from the joint.

Q49. [W2 | Independence] [MCQ] Answer: (B)

The support is the triangle $\{(x,y): x>0, y>0, x+y<1\}$, which is NOT a rectangle.

Non-rectangular support always implies dependence, even if the density appears to factorize.

If we try $f=24xy=(24x)(y)$ — the factorization fails because the domain constrains both variables jointly.

Key rule: if knowing X restricts the possible range of Y , they cannot be independent.

Q50. [W11 | Z-Test] [NAT] Answer: 2.5

$$Z = (x - \mu) / (\sigma / \sqrt{n}) = (205 - 200) / (20 / 10) = 5 / 2 = 2.5.$$

This means X is 2.5 standard errors above μ . Quite far from what H_0 predicts.

For two-tailed at $\alpha=0.05$: $|Z|=2.5 > 1.96$. Reject H_0 .

$$p\text{-value} = 2 \cdot P(Z > 2.5) = 2(0.0062) = 0.0124. \text{ Very significant.}$$